

Day 13 Hybrid Notes

Finding sides and angles with right-triangle trigonometry | Cloze + outline + active learning

Name:	Date:	Period:
-------	-------	---------

Learning Targets
I can label opposite, adjacent, and hypotenuse from a reference angle.
I can choose sine, cosine, or tangent for a right-triangle problem.
I can use a calculator to estimate missing sides and angles.

Part 1: Cloze Notes - Essential Ideas

1. Sine equals opposite over _____.
2. Cosine equals adjacent over _____.
3. Tangent equals opposite over _____.
4. SOH stands for sine = opposite / _____.
5. CAH stands for cosine = adjacent / _____.
6. TOA stands for tangent = opposite / _____.
7. The opposite and adjacent sides depend on the reference _____.

Part 2: Outline Notes - Big Picture A. Trigonometry compares side ratios in right _____. B. The reference angle tells which side is _____. C. Choosing a ratio depends on known and unknown _____. D. Calculator estimates should be checked for _____.	Connection to Prior Math 1. What algebra skill shows up today? _____ 2. What diagram or graph skill shows up today? _____ 3. What is one place this could appear outside math class? _____ _____
---	---

Most Important Reminder
Before solving, label the diagram, name the relationship, and check whether the answer makes sense.

Day 13 Hybrid Notes

Guided setup + active learning

Part 3: Guided Setup Practice

1. In a right triangle, angle $A = 35$ degrees, opposite = x , hypotenuse = 12. Set up the equation.

Setup/work: _____

Answer: _____

2. Which ratio uses adjacent and hypotenuse?

Setup/work: _____

Answer: _____

Part 4: Error Analysis

A student labels the adjacent side as the hypotenuse.
What went wrong?

Correct idea:

—

Before You Solve Checklist

☐ I labeled the diagram or graph.

☐ I chose the correct relationship or formula.

☐ I substituted values carefully.

☐ I checked whether my answer makes sense.

Part 5A: Draw It

Draw a diagram, graph, or labeled example that matches today's lesson.

Part 5B: Two-Sentence Summary

Sentence 1: Today I learned that

_____.

Sentence 2: One thing I need to remember when solving is

_____.

Part 5C: Application Problem

A ramp makes a 12-degree angle with the ground and has length 20 ft. Estimate the vertical rise.

Setup: _____

Answer: _____

Sentence answer: _____